# Identification and Classification of Cross-outs in Handwritten Documents

D7047E – Advanced Deep Learning
Project Proposal

**Group 14**
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# 1 Abstract

Handwritten text recognition (HTR) is a critical task in document analysis, yet it is often hindered by the presence of cross-outs, where words are marked with lines or scribbles. These artifacts obscure content and significantly affect recognition accuracy. This project develops a deep learning-based approach to automatically identify and classify seven types of synthetic cross-outs applied to the IAM dataset. We propose a multi-stage pipeline involving binary detection and multi-class classification. Furthermore, we evaluate model robustness through a custom-curated dataset of 50 images to ensure generalization across unseen handwriting styles.

# 2 Introduction

Handwritten text processing is a fundamental challenge in document analysis and historical manuscript digitization [1]. A recurring obstacle in these texts is the presence of cross-outs, which signify author edits but act as noise for automated recognition systems [1].

The primary objective of this project is to develop a deep learning-based approach to automatically identify and classify these artifacts. As specified in the project requirements, the technical goals are:

1. **Binary Classification:** Distinguishing between a clean word and a crossed-out word.
2. **Cross-Out Type Identification:** Categorizing the deletion into one of seven styles (Single-Line, Double-Line, Diagonal, Cross, Wave, Zig-zag, or Scratch).
3. **Small-Scale Experimentation:** Creating a custom dataset of 50 images to evaluate model performance and generalization on new data [1].

# 3 Literature Review

The identification of artifacts in handwritten documents has shifted from heuristic-based features to deep learning methodologies, our review suggests:

- **CNN-based Classification:** Architectures like ResNet and EfficientNet are widely used to classify noise in document images due to their ability to learn hierarchical spatial features.
- **Vision Transformers (ViT):** While ResNet is excellent for local texture, ViTs are increasingly cited for their ability to capture long-range spatial dependencies through self-attention - crucial for identifying patterns like diagonal or zig-zag cross-outs spanning across words.
- **Transfer Learning:** Fine-tuning models pre-trained on large-scale datasets has become standard practice to achieve high accuracy with limited labeled document data.

# 4 Method

## 4.1 Dataset and Structure

The project utilizes a modified version of the IAM Handwriting Database [2]. The dataset contains synthetic artifacts categorized into seven distinct styles as shown in Table 1.

Table 1: Synthesized Cross-out Categories and Visual Descriptions

| No. | Category | Visual Characteristics |
|---|---|---|
| 1 | Single-Line | A single horizontal stroke through the center of the word. |
| 2 | Double-Line | Two parallel horizontal strokes obscuring the text. |
| 3 | Diagonal | A single slanted stroke across the word. |
| 4 | Cross | Two intersecting diagonal strokes forming an 'X'. |
| 5 | Wave | A continuous undulating or curvy line. |
| 6 | Zig-zag | Sharp, alternating angular strokes. |
| 7 | Scratch | Dense, heavy scribbles that significantly obscure the characters. |

The data is pre-separated into `CLEAN`, `MIXED`, and individual style subdirectories to facilitate training. Ground truth transcriptions are maintained in `gt.txt` files within each data split [1].

## 4.2 Data Handling and Feature Focus

A key distinction in our methodology is the treatment of the handwritten content. This project is formulated strictly as a **visual artifact classification task** rather than a sequence-to-sequence transcription problem:

- **Feature Isolation:** The underlying semantic text is treated as background or "base" information. The models are trained to prioritize the extraction of stroke geometry (the artifacts) over character features.
- **Image Normalization:** To preserve the original aspect ratio and the frequency/amplitude of the cross-out patterns, images will be resized using **padding** rather than interpolation-based stretching.

## 4.3 Implementation Strategy

Our strategy focuses on a hierarchical approach to decompose the complexity of the task, moving from general detection to fine-grained classification:

- **Two-Stage Pipeline Formulation:** Rather than a single 8-class model, we will implement a two-stage logic to improve interpretability and robustness:
  1. **Stage 1 (Binary Detection):** A classifier to distinguish between "Clean" and "Crossed-out" word images.

2. **Stage 2 (Style Identification):** A specialized classifier triggered only for images identified as "Crossed-out" to categorize them into the seven specific styles.

- **Architectures:** We will evaluate and compare two advanced paradigms:
  - **CNN-based (ResNet):** Utilizing residual connections to preserve stroke geometry and texture details, serving as a baseline for feature extraction.
  - **Transformer-based (ViT):** Leveraging self-attention mechanisms to capture long-range spatial dependencies across the entire word image.
- **Transfer Learning:** We will employ models pre-trained on large-scale datasets (e.g., ImageNet) and fine-tune them on the IAM synthetic data to overcome the specific constraints of the handwritten domain.
- **Evaluation Strategy:** To ensure a rigorous analysis beyond simple accuracy, we will calculate **Precision, Recall, and F1-scores** for each category. Additionally, **Confusion Matrices** will be utilized to analyze inter-class similarity and identify which specific cross-out styles are most visually ambiguous to the model.
- **Generalization Testing (Custom 50):** To verify robustness, the team will generate a custom dataset of 50 images using diverse handwriting styles and physical ink-on-paper cross-outs to test the model beyond the synthetic training distribution.

At this stage, we are conducting a technical review of these architectures; comprehensive experimental results and hyperparameter configurations will be included in the final project report.

### 4.4 Repository and Version Control

All project source code, trained model weights, and the experimental dataset are maintained on GitHub.

**Project Repository:** https://github.com/chirag-gurumurthy/D7047E_ADL/tree/project

# 5 Deliverables

According to the project handout [1], Group 14 will provide:

1. **Source Code and Trained Model:** Complete codebase and final weights hosted on GitHub.
2. **Evaluation Script and Dataset:** Scripts to reproduce results and the custom 50-image experimental dataset.
3. **Final Presentation:** A 10-minute presentation covering methods, implementation, results, and learning outcomes.

# References

[1] Pathirage, G. and Barney Smith, E. *ADL Project: Document Analysis.*

[2] Marti, U. and Bunke, H. *The IAM-database: an English sentence database for offline handwriting recognition.*

[3] Gurumurthy, C. *D7047E_ADL Project Repository*. https://github.com/chirag-gurumurthy/D7047E_ADL/tree/project.